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CHEMICAL REGULATION OF RESPIRATION

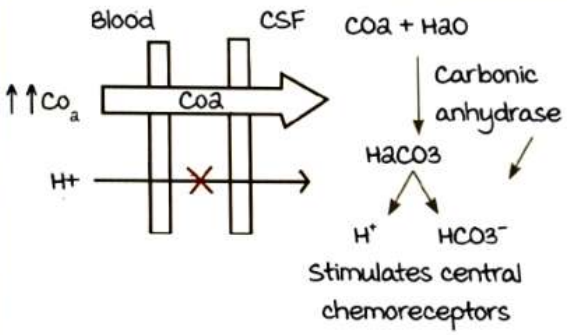
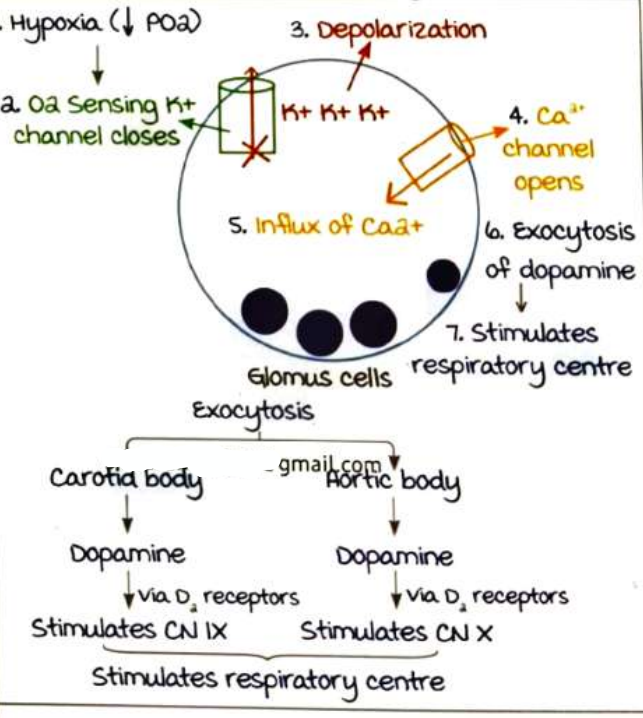
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Chemoreceptors

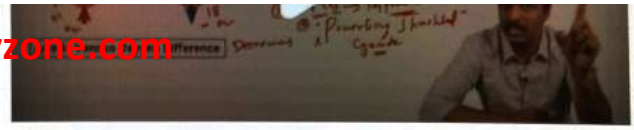
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Sense chemical changes during metabolism : $O_a\downarrow$, $CO_a\uparrow$, $H^+\uparrow$.

TYPES :

	Central	Peripheral	
Location	Ventral surface of medulla along floor of 4th ventricle	Neck  Carotid bodies Carotid bifurcation Aortic bodies (>2) Glomus Arch of aorta	
Types	–	Type I Glomus cells(O_a sensors)	Type II Glia like supporting cells
Stimulated by	↑ PCO2 (Blood), ↑ H+ (CSF), Hypotension	↓ O2 (Blood), ↑ H+ (Blood), Hypotension	
MOA	 Blood CSF $CO_2 + H_2O$ Carbonic anhydrase H_2CO_3 H^+ HCO_3^- $\uparrow\uparrow Co_a$ CO_2 H^+ Stimulates central chemoreceptors	 1. Hypoxia (↓ PO2) 2. O2 Sensing K+ channel closes 3. Depolarization K+ K+ K+ 4. Ca^{2+} channel opens 5. Influx of Ca^{2+} 6. Exocytosis of dopamine 7. Stimulates respiratory centre Glomus cells Exocytosis Carotid body Aortic body Dopamine Dopamine Via D_2 receptors Via D_2 receptors Stimulates CN IX Stimulates CN X Stimulates respiratory centre	
Applied aspect	Central respiratory depression/COPD O2 therapy C/I : • Hypoxia : Only stimulus for respiratory. • O2 therapy : Abolisher of respiratory drive.	Carotid body transplant tried as Rx for Parkinson's	
	Asphyxia → Central + peripheral chemo receptors activated → Chemoreflex → Hyperventilation → ↑Rate (↓PCO, ↓H+), ↑depth ↑PO2 • Fall in Po2 (Hypoxia). • Rise in PCO2 (Hypercarbia). • Rise in H+ (Acidosis).		

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Hypoxia

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O2 Transport

- Dissolved form
- O_a – Hb (Reduced O_a content)

TYPES :

	Hypoxic	Anemic	Stagnant/ischemic	Histotoxic
Seen in	• High altitude. • COPD.	• Anemia. • Carbon monoxide (CO) poisoning.	-	Cyanide poisoning
Mechanism of Action	↓ dissolved form ↓ ↓ PO2 (Hypoxia) ↓ Stimulates peripheral chemoreceptor Rx : O2 therapy	• ↓ O2 – Hb. • In CO poisoning, ↑ affinity of CO to Hb ↓ Defective O_a release to tissue ↓ Left shift on oxygen dissociation curve	Stasis of venous blood ↓ ↑ O2 utilization at tissue ↓ ↑ Arteriovenous oxygen (AVO2) difference	↑ cyanide ↓ Inhibits cytochrome oxidase ↓ Defective O2 utilization ↓ ↓ AVO2 difference
Peripheral Chemo-Receptor	Stimulated	Not stimulated	Not stimulated	Stimulated (Cyanide : most powerful stimulator)

Feedback